

Andrea Grillo

EPFL MASTER'S STUDENT IN ROBOTICS

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Summary

As a dedicated Robotics student with a foundation in Computer Science, I'm interested in the **deployment of complex algorithms on resource-constrained edge platforms**. I aim to explore reduced-precision computation and alternative computing platforms within the context of robotics and control algorithms.

Education

Dartmouth College

Hanover NH, USA

VISITING MASTERS THESIS RESEARCHER

Sept. 2025 - Mar. 2026

- Research on the algorithmic adaptation of ADMM-based solvers for linear MPC for neuromorphic chips and FPGAs. Emulation of neuromorphic chip on GPU, FPGA implementation and deployment on custom PCB on Crazyflie quadcopter.
- Supervisor: Prof. Brian Plancher, A2R Lab.

EPFL - École Polytechnique Fédérale de Lausanne

Lausanne, SWITZERLAND

MASTER'S DEGREE IN ROBOTICS

Sept. 2023 - exp. grad. Apr. 2026

- Relevant courses: Model Predictive Control, Embedded System Design, System Programming for SOCs, Distributed Algorithms.
- Teaching assistant for courses: Basics of Mobile Robotics, Basics of Robotics for Manipulation. Grade average: 5.83/6.

POLITO - Politecnico di Torino

Torino, ITALY

BACHELOR'S DEGREE IN COMPUTER ENGINEERING

Aug. 2020 - Jul. 2023

- Final grade: 110/110 cum laude. Intraprendenti Project: excellence program for top-level students.
- Camplus Bernini: INPS fellowship for living in a College of Merit during my studies.

Professional Experience

Hydromea

Renens, SWITZERLAND

INTERNSHIP - UNDERWATER OPTICAL COMMUNICATION ENGINEER

Feb. 2025 - Jul. 2025

- New firmware version of optical underwater modem. Owned firmware quality assurance. Designed and automated production test infrastructure for manufacturing. Customer technical support. Coordinated and executed firmware release to production.
- Embedded systems, C, Python, Git, CI/CD, Docker.

BIOROB Laboratory

Lausanne, SWITZERLAND

EPFL SEMESTER PROJECT

Sept. 2024 - Jan. 2025

- Implementation of high frequency, low latency data acquisition system for a robotic platform.
- ROS2, C++, RS485.

Automatic Control Laboratory - Predict Lab

Lausanne, SWITZERLAND

RESEARCH ASSISTANT AND SEMESTER PROJECT

Feb. 2024 - present

- Distributed Model Predictive Control for miniature hovercrafts. Implementation of the software framework for the deployment of distributed MPC on small hovercraft aiming at multi-robot formation control. This work was accepted to ICRA 2025.
- ROS2, Optimization, C++, MPC.

Swiss Solar Boat

Lausanne, SWITZERLAND

EPFL SEMESTER PROJECT

Sept. 2023 - Feb. 2024

- Design of a framework for the software architecture of the new prototype boat.
- ROS2, CANBUS communication, Python/C++ programming, Docker containers.

Pic4Ser - Politecnico di Torino

Torino, ITALY

STUDENT INTERNSHIP

Feb. 2023 - Jul. 2023

- Design, project and realization of an obstacle-avoidance sensor systems for an existent robotic platform.
- STM32 microcontroller PCB design and programming, CANBUS communication.

Team ISAAC - Politecnico di Torino

Torino, ITALY

TEAM LEADER

Nov. 2020 - Jul. 2023

- Leadership and management of a student team. Design and implementation of a biomimetic modular snake-like robot.
- Microcontroller programming, control systems, network management, software development management.

Openscuola Project - Politecnico di Torino

Torino, ITALY

SPECIALIZED TECHNICIAN - STUDENT INTERNSHIP

Oct. 2021 - Jul. 2023

- Public schools IT infrastructure installation and maintenance using recycled hardware and open-source software.
- <https://linux.studenti.polito.it>

Publications

2025

- *Stomberg, G., Schwan, R., Grillo, A., Jones, C.N. and Faulwasser, T.* - "Cooperative distributed model predictive control for embedded systems: experiments with hovercraft formations." in Proceedings of the IEEE International Conference on Robotics and Automation (ICRA) 2025.
Available at: <https://arxiv.org/abs/2409.13334>

Languages

English, C1 - IELTS certificate: Overall Band Score 8.0

French, B2

Spanish, C1 - CELU certificate: Advanced Level - Mention of Excellence

Italian, Mother Tongue

Competences

PROGRAMMING

- Excellent C/C++ for embedded systems and low-level peripheral interfacing.
- Proficient in Python, Java, MIPS/OpenRISC assembly.
- Experience in MATLAB for data analysis and simulation.
- Version control expertise with Git (GitHub/GitLab).

ROBOTICS AND CONTROL SYSTEMS

- Proficient in ROS/ROS2 for embedded systems.
- Control systems: PID, MPC, distributed DMPC, optimizers for MPC.
- Experience in robotics hardware: microcontrollers, motor drivers, sensors, power systems.
- Robot design: sensor/actuator selection, transmission sizing, kinematic computation.

IT AND NETWORKING

- Expertise in GNU/Linux (Debian, Ubuntu) for both server and embedded environments.
- Proficient in virtualization using Docker.
- Experience in networking: VPNs (Wireguard), routing, firewall management.

ELECTRONICS AND EMBEDDED SYSTEMS

- Analog and digital circuit design: OpAmps, DC/DC converters, FPGA, ADC/DAC.
- Proficient in communication protocols: I2C, SPI, RS-485, CANBUS.
- PCB design for microcontroller and FPGA systems (STM32, Artix-7).

OTHER SKILLS

- Strong manual skills for assembly and repairs (mechanical/electrical systems).
- Leadership and teamwork management (GitHub, GitLab CI/CD, Slack, Mattermost, ClickUp, Trello, Notion).